Disconnected strings in two Coxeter families

Human Compression project: F006 recovery

A string C-marking is an ordered list of nonidentity involutions generating a finite group, with nonadjacent entries commuting and with $G_I \cap G_J = G_{I\cap J}$ for every two sets of positions, where G_I is generated by the entries in I. We allow adjacent entries to commute. Such a marking is called disconnected when at least one adjacent product has order two. These conventions are the literal ones in Chen and Rowley [1]; their discussion of B_n expressly includes disconnected strings.

Their Theorem 3(i) and Table 1 require a qualification under this convention. The two cases addressed here are $I_2(m)$ and H_3. The eight connected H_3 markings in the source table are recovered, but there are also three rank-four classes. This note does not certify the other exceptional groups or the D_n results. No new open-conjecture answer is asserted.

1 The dihedral group

Theorem 1. *Let $m \geq 3$ and let $D = \langle r, s \mid r^m = s^2 = 1,\ srs = r^{-1}\rangle$. With disconnected strings allowed,*

$$r_{\max}(D) = \begin{cases} 3, & m \equiv 2 \pmod 4, \\ 2, & \textit{otherwise.} \end{cases}$$

Every rank-three string C-marking of D is disconnected.

Proof. The involutions are the reflections $r^i s$ and, when m is even, the central involution $z = r^{m/2}$. Two distinct reflections commute precisely when their product is z. The intersection property makes every generator independent of the others, and in particular makes all generators distinct.

There is no marking of rank at least four. If z occurs, at least three other entries are reflections; two of their positions are nonadjacent. They commute, and their product is z, contrary to independence. If z does not occur, the first reflection commutes with both the third and the fourth. A reflection has at most one distinct commuting reflection, again a contradiction.

Suppose a rank-three marking consists entirely of reflections. Its end entries commute, so m is even. Change the name of the first reflection to s; the last is zs, and the middle has the form $r^t s$. Generation requires $\gcd(t, m/2) = 1$. If $m/2$ is even, then t is odd, so $\gcd(t, m) = 1$ and the first two entries already generate D. If $m/2$ is odd, either t is odd, with the same consequence, or $m/2 - t$ is odd and coprime to m, in which case the last two entries generate D. Every case contradicts independence.

A rank-three marking must therefore consist of z and two reflections a, b. Put $L = \langle a, b\rangle$. Independence says $z \notin L$. If $h = |ab|$, the rotation subgroup of L has order h and contains z exactly when h is even. Thus h is odd, and generation of D gives $m = 2h$. The central entry cannot be in the middle, since the two end reflections would then commute and generate it. The marking is consequently disconnected.

Conversely, if $m = 2h$ with odd $h \geq 3$, then $D = \langle z\rangle \times \langle s, r^2 s\rangle$. The second factor is the dihedral group of order $2h$. Its two distinct generating reflections have the rank-two intersection property. For the marking $(z, s, r^2 s)$, every subgroup generated by a subset is the direct product of its two coordinate subgroups, so the intersection property follows coordinate by coordinate. Its type is $\{2, h\}$. The usual pair (s, rs) gives rank two in every case. □

For example $I_2(6)$ has the disconnected rank-three marking of type $\{2,3\}$. The rank-two assertion is valid when connected strings alone are intended.

2 The icosahedral group

We retain the classical finite Coxeter order $|W(H_3)| = 120$. Its identification with the following permutation group is verified explicitly. Set

$$a = (1\ 2)(3\ 4), \quad b = (1\ 2)(3\ 5), \quad c = (1\ 3)(2\ 4), \quad z = (6\ 7).$$

The products ab, bc, ac have orders $3, 5, 2$, respectively. Hence $\langle a, b, c\rangle \leq A_5$ has subgroups of orders $6, 10, 4$; its order is divisible by their least common multiple, sixty, and it is A_5. The product abc has order five. Therefore the Coxeter triple (za, zb, zc), of type $\{3, 5\}$, generates $A_5 \times \langle z\rangle$: its product has fifth power z. The resulting surjection from $W(H_3)$ is an isomorphism by the retained order. No isomorphism with a group of the same order alone is presumed.

Lemma 2. *The marking (z, a, b, c) is a rank-four string C-marking of $W(H_3)$, of type $\{2, 3, 5\}$.*

Proof. The three rank-two subgroups for (a, b, c) have orders $6, 10, 4$. Their pairwise intersections have orders dividing two and contain the respective common generator. Thus each intersection is precisely the common subgroup. Together with their cyclic subgroups and the trivial subgroup this checks the intersection property for all eight subsets. The end involutions a, c commute. Finally adjoining z as a separate end factor preserves every intersection coordinate by coordinate, as in the preceding theorem. □

Theorem 3 (Complete small census)**.** *Up to group automorphisms and reversal, the string C-markings of $W(H_3)$ have the following counts. Parentheses give the number of self-dual classes.*

	rank 3	*rank* 4	*rank* ≥ 5
connected	8 (1)	0	0
disconnected	0	3 (0)	0

In particular its maximum rank is four with both conventions included, and three when connected strings alone are considered.

Computational proof with the complete finite search specified. Use the 60 even permutations of $\{1, \ldots, 5\}$, each independently fixing or exchanging $6, 7$, to obtain 120 elements. Multiplication is ordinary permutation composition. There are 31 nonidentity involutions. The program `check_h3.py` constructs this whole multiplication table; it uses no group classification or canonical-label library.

It grows every ordered prefix of involutions. When appending x, it requires x to lie outside the subgroup already generated, checks its commutation with all earlier nonadjacent entries, and tests the intersection property for every pair of subsets of the enlarged prefix. Subgroups are computed by repeatedly multiplying known elements by the specified generators until no new element remains. These are exact finite set operations. Every prefix of a string C-marking has the intersection property and independent generators, so no possible marking is lost by these rejections. When the generated subgroup is the whole group, the marking is recorded; it cannot be extended independently. Each proper extension at least doubles the subgroup order. Thus any marking has $2^r \leq 120$, and it suffices to search through rank six. The numbers of valid prefixes reached at ranks one through six are

$$31, \quad 930, \quad 3720, \quad 720, \quad 0, \quad 0.$$

For completeness of the equivalence calculation, every automorphism of $A_5 \times C_2$ preserves its derived subgroup A_5 and its centre C_2. Here A_5 is perfect because a commutator of two double

transpositions is a three-cycle, and conjugate three-cycles generate A_5; its centre is trivial. Hence the automorphism group is $\mathrm{Aut}(A_5)$. There are five Sylow two subgroups of A_5, one Klein four subgroup fixing each point. An automorphism permutes these five subgroups faithfully: if it fixes them all, conjugation by g and by its image have the same permutation on them for every g, and the natural A_5 action is faithful. Thus $\mathrm{Aut}(A_5)$ embeds in S_5. Conversely conjugation by S_5 supplies 120 distinct automorphisms. These are all of them.

The program therefore takes the least tuple among these 120 conjugates and their reversals. A class is self-dual exactly when its reversal is among its 120 conjugates without reversal. It tests connectedness by whether all adjacent product orders exceed two. The full representative lists and the complete execution give the displayed counts. In particular, no truncated search or sampled list supplies the zero columns. Correct execution of the supplied Python integer and permutation program remains the explicit computational premise; this is not a Lean formalization. □

Attribution and scope. The source's definition, rank claims and eight connected H_3 classes are those of Chen and Rowley. The dihedral structure, direct-product intersection argument, classical order of H_3, and automorphism description of A_5 are standard inputs or elementary arguments given here. The present work makes the disconnected boundary explicit and gives an independent complete small enumeration. This is a correction to the literal scope, not a claim that the source's connected census is wrong. The other five exceptional census rows, the D_n constructions and all whole-manuscript admission obligations remain outstanding. No priority claim or conjecture credit is attached to this bounded result.

References

[1] M. H. W. Chen and P. Rowley, *Abstract Regular Polytopes of Finite Irreducible Coxeter Groups*, Electronic Journal of Combinatorics **32**(4) (2025), #P4.18, doi:10.37236/13763, fixed thirteen-page published edition, especially Theorem 3 and Table 1.